Flows of energy through ecosystems

The word *trophic* refers to nutrition and *trophic levels* are the different levels through which nutrition flows by way of the food pyramid or food chain. Organic matter, called *biomass* (*bios* living, *mass* a quantity of matter), travels through the food chain starting at the level of primary producer and peaking at the top carnivore.

1 Explain what is meant by a *food chain.*

The amount of available biomass (and energy that comes with it) is reduced at each level above the primary producer; in other words, as it travels through the food chain. This happens because some of the energy is incorporated into the organism that eats it. For this reason, a food pyramid tends to reach a maximum of five levels, usually fewer.

At the most basic level we find the *primary producer.* Primary producers can create their own food by converting the sun's energy via a process called *photosynthesis.* Primary producers are found on land (terrestrial) and in water (aquatic).

2 Give an example of *a terrestrial primary producer.*

3 Give an example of an *aquatic primary producer.*

The trophic levels above the primary producer level are collectively called *consumers* because they obtain nutrients and energy by consuming other organisms. However, consumers can be classified in several ways. At the second level of the food pyramid we find the *herbivores* (plant-eaters) also called grazers. These include all species that do not eat meat, including plant-eating insects. The levels above the herbivores are the *carnivores* (meat-eaters) and there can be several ranks.

4 Name two native Australian herbivores.

The term *carnivore* usually refers to the first meat-eating level, but since carnivores can also be eaten, a *secondary carnivore* and a *tertiary carnivore* (sometimes called *super carnivore*) are included in some classifications. For example, a snake eats other animals but owls can also eat this meat-eater. In the case of two fish-eating animals, the shark and the seal (excluding leopard seal), both are considered carnivores. However, since the shark would just as easily eat the seal as eat a fish, but the seal would not eat a shark, the shark would be the secondary carnivore, placed above the seal on the pyramid. There are not many species of tertiary carnivore. They include humans, killer whales and leopard seals. Tertiary carnivores are seldom hunted as prey.

The Great skua (*Stercorarius skua*) and the Australian dingo (*Canis sp.*) provide good examples of the complexities of the food web and how an organism can be classified in more than one position. The Great skua is a large predatory bird, generally found in the colder parts of the Northern Hemisphere (ie Iceland, Scotland), Antarctic and southern South America. The Australian dingo is a wild dog that prefers to feed as a carnivore, however, its success lies in its flexibility when it comes to diet. Dingoes have been seen eating fish, seaweed and other plant matter when meat is unavailable.

5 How would you classify the skua when it hunts fish, like other fish-eating birds, but also hunts fish-eating birds? Explain your answer.

6 How would you classify a dingo on the food pyramid when it eats meat and when it eats plant matter? Explain your answer.

7 Give the general name for an animal that eats both plant and animal matter.

8 Would you consider it accurate to classify the dingo as the name given in your answer to question 7. Explain your answer.

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